Museum Monitoring

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Abstract
Museums increasingly deploy environmental sensors to support the conservation of cultural heritage objects. While sensing technologies are widely available, the resulting data is typically siloed within proprietary dashboards and local infrastructures, limiting its reuse, analysis, and controlled sharing—particularly in cross-institutional scenarios such as object loans. This paper reports on MuMo (Museum Monitoring), a three-year applied research project that explored how dataspace principles can be applied to environmental monitoring in real museum settings. Rather than replacing existing systems, MuMo integrates semantic data modeling, Linked Data Event Streams, and Solid-based data governance with a legacy monitoring dashboard. This enables decentralized data publication, selective cross-institutional access, and client-side aggregation across multiple independent deployments. We present the system design and describe how it is used in practice through a set of in-use scenarios, including longitudinal analysis of environmental conditions, controlled data sharing during loans, and multi-source data integration. The paper further reflects on design trade-offs, particularly the choice of group-based access control aligned with existing curatorial workflows, despite the technical feasibility of finer-grained authorization. The contribution of this work lies in the lessons learned from deploying a dataspace-oriented architecture under real institutional constraints. These insights are relevant to a broad range of Digital Humanities projects involving long-running data collection, distributed governance, and interoperable yet non-centralized infrastructures.

Keywords
Digital Humanities, Dataspaces, Solid, LDES, Museum Monitoring

1. Introduction & Motivation

Museums increasingly rely on digital infrastructures to support the conservation, documentation, and circulation of cultural heritage objects. Among these infrastructures, long-term environmental monitoring plays a crucial role: parameters such as temperature, humidity, and light exposure directly affect the preservation of artworks and are often contractually regulated, particularly in the context of inter-museum loans. As a result, monitoring data must not only be collected reliably, but also interpreted, analyzed, and—under specific conditions—shared across institutional boundaries.

Conservation practice increasingly frames environmental control in terms of risk management and tolerable fluctuation ranges rather than a single “ideal” climate, reinforcing the operational importance of continuous monitoring and traceable reporting [14].

A wide range of sensing technologies for environmental monitoring already exists and is actively used in museum practice, including long-running wireless sensor network (WSN) deployments in museum buildings [15]. However, these systems typically operate as stand-alone solutions, tightly coupled to proprietary dashboards or local infrastructures. As a consequence, monitoring data remains siloed: it is difficult to combine measurements across installations, to align them with other data sources, or to selectively share them with external partners. This fragmentation limits the analytical value of the data and complicates collaboration between institutions.

These challenges are exacerbated by the realities of museum infrastructures. Monitoring systems are often embedded in long-lived legacy dashboards that were designed for local use,

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provide limited access control mechanisms, and do not support cross-institutional data reuse. At the same time, museums operate within strong organizational and governance constraints: institutional autonomy must be preserved, technical solutions must remain manageable by non-technical staff, and existing systems cannot easily be replaced.

The limitations of siloed monitoring become particularly visible in loan scenarios, where lending institutions require access to environmental data produced by borrowing museums, but only for a specific subset of sensors and a limited time period. Existing practices often rely on ad-hoc data exports or manual reporting, leading to delays, duplication, and reduced transparency in conservation oversight.

Recent work has similarly highlighted how the increasing digitization of museum collections exposes structural limitations in legacy monitoring and documentation infrastructures, particularly with respect to long-term traceability, institutional accountability, and resilience against security incidents [16].

This paper reports on MuMo (Museum Monitoring), a three-year applied research project that explored how dataspace principles can be applied to environmental monitoring in real museum settings, namely prioritizing incremental integration, heterogeneous sources, and governance [7, 8]. Rather than proposing new sensing technologies or replacing existing platforms, MuMo focused on breaking down data silos by integrating semantic technologies, decentralized data publication, and access control mechanisms with existing systems and workflows. The contribution of this paper lies in documenting how these design choices functioned in practice, which trade-offs were made, and what lessons can be drawn for the broader Digital Humanities community.

2. Context of Use

The MuMo (Museum Monitoring) project was a three-year applied research project aimed at supporting museums in the long-term monitoring of environmental conditions around collection objects. Many museums—especially smaller institutions—lack continuous insight into parameters such as temperature, humidity, or light exposure, despite these being critical for conservation and often contractually relevant during object loans.

2.1. Operational Setting

MuMo was deployed in real museum environments and focused on low-maintenance, long-running installations. Custom ultra-low-power sensing hardware was developed to operate for approximately one year without recharging. Sensors were placed near artworks or storage locations and continuously measured environmental parameters. Due to the physical and organizational constraints of museum spaces, the system had to function with minimal intervention once deployed.

Measurements were transmitted via The Things Network and ingested into an existing (legacy) monitoring dashboard. This dashboard became the primary operational interface for museum staff and therefore strongly shaped how data could be accessed, interpreted, and shared.

2.2. Legacy Dashboard Constraints

The existing dashboard provided:

- User and group management, including recursively defined groups
- Node management corresponding to deployed sensors
- Basic visualization through simple time-series graphs

Access control was **group-based and coarse-grained**, sufficient for internal monitoring within a single institution but not designed for cross-institutional collaboration. Importantly, the dashboard could not be replaced or fundamentally re-engineered within the scope of the project, requiring MuMo to operate within these constraints.

2.3. Cross-Institutional Access and Loans

A central real-world use case was **the monitoring of artworks during loans between museums**. Lending institutions often require insight into the environmental conditions under which an object is kept while on loan, without being granted broader access to the borrowing museum's infrastructure or internal data.

In practice, this requirement translates into:

- access limited to a subset of sensors,
- bounded by the duration of the loan,
- manageable by non-technical museum staff.

Although often described as "fine-grained access control," the actual operational need is **group-level authorization**: sensors associated with a specific loan can be grouped, and access to that group can be granted temporarily to another institution.

2.4. Requirements for Data Sharing

These scenarios impose several practical requirements:

- Data must be shareable across **institutional boundaries**.
- Access must respect existing group-based authorization practices.
- Users must be able to access data originating from multiple museum setups.
- Identity management cannot assume a single central authority.

These requirements motivated an architecture in which monitoring data could be published, discovered, and accessed across organizational boundaries, while remaining compatible with legacy systems and established museum workflows.

3. System and Approach Overview

3.1. Dataspace Architecture Based on Solid

MuMo adopts a **dataspace-oriented architecture** based on Solid, a decentralized data platform designed to give data owners control over their data rather than centralizing it within applications [13, 22, 2]. Solid is explicitly designed as a decentralization framework in which data storage, identity, and access control are decoupled from individual applications, enabling data to be reused by multiple clients and actors without funneling all data through a centralized service [13, 22].

This approach aligns with museum practice: institutions are unwilling to relinquish control over their monitoring infrastructure, yet must be able to selectively share data during collaborations such as object loans. Solid provides a conceptual and technical framework in which identity, authorization [3], and data location are decoupled from any single application, explicitly aiming to counter the concentration of data within centralized platforms by restoring data ownership to the producing parties. .

While some recent research emphasizes cryptographic immutability and tamper resistance through tightly coupled sensor infrastructures, hardware-based attestation, and private distributed ledgers, MuMo instead prioritizes institutional autonomy and operational feasibility by enabling decentralized data publication and access control aligned with existing museum workflows [16].

In MuMo, Solid Pods act as **institutional data endpoints** rather than user-centric storage, enabling long-lived publication of monitoring data that remains independently governed.

3.2. Continuous Data Publication with Linked Data Event Streams

Environmental monitoring produces **continuous, append-only data** that grows over time and is rarely modified retroactively. To reflect this, MuMo publishes monitoring data using **Linked Data Event Streams (LDES)** [21, 5].

LDES is increasingly adopted as a practical pattern for interoperable publication of evolving datasets, enabling replication and synchronization across organizational boundaries without centralized query services [5].

Prior work has demonstrated the feasibility of storing and consuming LDES event sources within the Solid ecosystem, supporting the compatibility of event-stream publication with Solid-based access control [17].

LDES enables consumers to:

- retrieve historical data incrementally,
- stay synchronized with newly produced observations,
- avoid repeated querying of centralized services.

This publication model proved particularly suitable in a cross-institutional setting, as it allows data consumers to process only the subsets of data they are authorized to access, without requiring the data provider to offer tailored query endpoints [21].

3.3. Semantic Representation of Sensors and Observations

All data managed by the legacy dashboard is transformed into a semantic representation, including both **environmental observations** and **sensor configuration metadata**. This separation reflects two different kinds of change:

- observations evolve continuously over time,
- sensor configurations evolve discretely when sensors are moved or reconfigured.

The semantic representation builds on established models for describing sensors and observations, most notably the W3C Semantic Sensor Network ontology (SSN/SOSA) [4, 9], which provides a shared conceptual framework for sensors, observations, and their relationships.

SN/SOSA was selected in particular for its explicit separation between sensors, observations, and the features of interest being observed, enabling sensor redeployment and contextual change to be represented without altering historical observations [4].

In MuMo, these concepts are instantiated using OSLO vocabularies [1], a Flemish Linked Data standardization framework that profiles and reuses international semantic standards for cross-organizational data exchange.

Sensor metadata is therefore published as a **versioned event stream**, allowing consumers to reconstruct the context in which observations were produced. Group membership and location are encoded explicitly in the semantic descriptions, enabling downstream systems to reason about authorization and interpretation without consulting the legacy dashboard.

3.4. Group-Based Access Control as a Practical Design Choice

Access control in MuMo deliberately mirrors the **group-based authorization model** already in use in the legacy dashboard. Rather than introducing fine-grained authorization at the level of individual observations, access is granted at the level of sensor groups.

At the Solid protocol level, this aligns with authorization mechanisms such as Web Access Control (WAC), which allow servers to enforce access rules on resources and their associated representations [2, 3].

This choice was guided by museum practice: for loan scenarios, institutions require access to all monitoring data related to a specific object or location over a defined period. Prior work on access control has repeatedly highlighted that fine-grained authorization models, while expressive, introduce significant configuration and usability complexity in operational settings, particularly for non-technical users [10]. Group-level access was therefore found to be both sufficient and manageable, avoiding complexity that would hinder adoption.

Authorization policies defined in the legacy system are reflected in access constraints on published data, allowing cross-institutional sharing without introducing a new access management interface.

Although the underlying fragmentation strategy and event-based publication model would technically allow finer-grained access control (e.g., at the level of individual days or measurement types), MuMo intentionally limits authorization to group-level permissions in order to remain aligned with the configuration mechanisms and functional requirements of the legacy dashboard.

3.5. Decentralized Consumption and Aggregation

Because each MuMo deployment publishes its data independently, consumers may need to combine data from multiple sources. MuMo supports this through a client-side dashboard that retrieves sensor descriptions first, determines the user's authorized scope, and then incrementally consumes the relevant observation streams.

This approach avoids centralized aggregation while still enabling a unified user experience, reinforcing the dataspace principle that **integration occurs at the point of use**.

4. In-Use Scenarios

This section illustrates how the MuMo dataspace architecture is used in practice by museum professionals to analyze, share, and contextualize environmental monitoring data.

4.1. Scenario 1: Analyzing Environmental Conditions Over Time

The primary use of the advanced dashboard is to support museum staff in assessing the long-term "health" of collection objects by analyzing environmental conditions over time. Users interact with the system through a web-based dashboard that allows them to filter and combine data based on:

- location (group),
- sensor or node,
- type of measurement (e.g., temperature, humidity),
- time constraints.

Using these filters, users can construct queries that follow an artwork throughout its lifecycle. For example, a curator can analyze how environmental conditions evolved while an object was stored in one room, then exhibited in another, and later placed in temporary storage. Because sensor configuration changes are published as versioned semantic descriptions, the system can correctly associate observations with their deployment context at each point in time.

A key practical benefit of the Linked Data Event Streams (LDES) publication model is that **data filtering occurs before retrieval**. Rather than fetching all historical measurements and filtering client-side, the dashboard retrieves only the relevant fragments based on semantic relations and temporal constraints. For instance, when analyzing conditions in a specific year, only the corresponding fragments are accessed, avoiding unnecessary data transfer and improving responsiveness.

This incremental and selective access is essential for long-running monitoring installations, where datasets grow continuously and may span multiple years.

4.2. Scenario 2: Cross-Institutional Access During Loans

A second, critical scenario concerns the monitoring of artworks during loans between museums. Lending institutions typically require access to environmental data from the borrowing museum to ensure that conservation conditions meet agreed standards, while borrowing institutions must retain control over their broader monitoring infrastructure.

In MuMo, this scenario is supported through **group-based access control aligned with Solid identities**. For a loan, the borrowing museum creates a dedicated group in the legacy dashboard and associates the relevant sensors with that group. Access to this group is then granted to specific Web-based identifiers belonging to the lending institution.

Because access control is enforced at the level of published data fragments, external users can authenticate using their own WebIDs and access only the data streams corresponding to the loan group. No centralized user management or data replication is required. Once the loan period ends, access can be revoked by removing the external WebIDs from the group.

This approach enables temporary, fine-grained sharing of monitoring data across institutional boundaries while remaining manageable for museum staff and compatible with existing workflows.

4.3. Scenario 3: Integrating Multiple Data Sources

In addition to supporting analysis within a single monitoring deployment, the advanced dashboard demonstrates the ability to **combine data from multiple independent MuMo data sources**. Each MuMo deployment publishes its monitoring data and sensor descriptions independently, yet follows the same semantic representation and event-based publication model.

In practice, this allows users to access and analyze data originating from different museum setups within a single interface. For example, a user may compare environmental conditions across multiple exhibition spaces or institutions, provided they have the appropriate access rights. Because sensor metadata and observations are published as Linked Data Event Streams, the dashboard can discover available sensors, determine authorization, and incrementally retrieve data from multiple sources without requiring centralized aggregation.

Beyond this demonstrated functionality, the same mechanisms also enable the **conceptual integration of external data sources** that are not part of the MuMo project, such as weather station measurements. Since both sensor metadata and observations are modeled using shared semantic standards, incorporating additional event streams would not require changes to the underlying architecture. While such external integrations have not yet been deployed, they directly informed the design of the system and illustrate how the dataspace approach supports extensibility and reuse.

4.4. Generalizing the scenarios beyond monitoring systems

While the scenarios above focus on environmental monitoring data, they also reveal a more general pattern regarding the role of contextual information in enabling downstream analysis. They rely on contextual information such as temporal validity, object location, and sensor deployment. In MuMo, this information was obtained from the monitoring infrastructure and its associated configuration data. However, the underlying mechanism illustrated by these scenarios does not depend on the specific system from which such context originates. In practice, museums already maintain similar forms of structured contextual information in a variety of collection management and repository systems, independent of environmental monitoring.

Contemporary collection management systems and digital repositories often manage object metadata, temporal associations, and organizational context, and may support structured export mechanisms. Examples include repository platforms such as Omeka-S or digital repository services such as D-RaaS [19].

When such contextual information is made available in a structured and time-aware form, it becomes possible to derive queries that are not explicitly supported by the originating CMS.

Instead, these queries emerge from the combination of temporal relations, location changes, and object associations at the point of use.

MuMo demonstrates this principle in the domain of environmental monitoring, but the same mechanism can be applied more broadly to contextual data maintained by other institutional systems. This suggests that dataspace-oriented approaches can unlock additional analytical value from existing CMS infrastructures without requiring their replacement or redesign.

5. Observations and Lessons Learned

The in-use scenarios presented above highlight how specific architectural and modeling decisions shaped the practical use of MuMo in museum monitoring contexts. Rather than evaluating individual technologies in isolation, this section reflects on how these choices interacted with real-world constraints and practices.

5.1. Solid as an Institutional Boundary Mechanism

Across all scenarios, Solid played a central role in enabling data sharing without centralization. By treating Solid Pods as **institution-level data endpoints** rather than user-centric storage, MuMo aligned naturally with museum governance structures. Institutions retain control over their monitoring data while still enabling external access when required.

This was particularly evident in the loan scenario (Scenario 2), where Solid identities (WebIDs) allowed users from different institutions to authenticate without relying on a shared user database. Access could be granted and revoked by modifying group membership, supporting temporary collaborations without introducing new identity management workflows.

Lesson learned: Solid is most effective in this context when used to represent **organizational boundaries and responsibilities**, rather than as a personal data store.

5.2. Linked Data Event Streams for Scalable Analysis

The use of Linked Data Event Streams proved essential for handling the continuous and growing nature of monitoring data. In the analysis scenario (Scenario 1), LDES enabled the dashboard to retrieve only the relevant subsets of data based on semantic relations and temporal constraints, avoiding the need to fetch complete datasets and filter client-side.

This incremental access model also facilitated multi-source analysis (Scenario 3), where data from multiple independent MuMo deployments could be consumed and combined without requiring centralized aggregation or bespoke query endpoints.

Lesson learned: Event-based publication is particularly well suited for long-running Digital Humanities data collection, where datasets evolve continuously and selective access is more important than expressive querying.

5.3. Group-Based Access Control as a Deliberate Design Choice

MuMo deliberately adopts **group-level access control**, even though finer-grained authorization is technically feasible within the chosen architecture. The use of Linked Data Event Streams combined with a hierarchical fragmentation strategy makes it possible to enforce access constraints at different levels of granularity, such as individual days or, with minor adaptations to the publication pipeline, per measurement type.

In practice, these options were not activated. The legacy dashboard that remains authoritative for configuration and access management does not provide mechanisms to express such fine-grained permissions. As a result, only those authorization concepts that are meaningful and manageable within existing museum workflows were reflected in the generated access control configurations.

This decision was particularly evident in the loan scenario, where group-level access proved sufficient to grant lending institutions visibility into relevant monitoring data without exposing unrelated information. Introducing finer-grained access control would have increased technical complexity without corresponding benefits for end users.

Lesson learned: In applied Digital Humanities settings, the appropriate granularity of access control is determined less by technical capability than by the expressive power of existing organizational tools and practices. Designing for feasible access control is often more valuable than designing for maximal access control.

5.4. Semantic Modeling as an Enabler of Integration

Semantic representations based on shared observation models enabled MuMo to separate **data production** from **data consumption**. Sensor descriptions, deployment context, and observations could be published independently, versioned over time, and interpreted consistently across systems.

This separation was critical in Scenario 1, where sensor relocations needed to be reflected correctly in longitudinal analyses, and in Scenario 3, where data from multiple sources could be combined as long as they adhered to the same semantic structures.

Lesson learned: The value of semantic modeling in practice lies less in expressiveness and more in its ability to support evolution, reuse, and alignment across independently managed systems.

5.5. Dataspace Integration at the Point of Use

Finally, the combination of Solid, LDES, semantic modeling, and client-side aggregation reflects a broader dataspace principle: **integration happens at the point of use**, not through centralized infrastructure.

Similar architectural patterns can be observed in other domain-specific data space initiatives, such as the Flanders Smart Data Space, where semantic standardization and Linked Data Event Streams are combined to support decentralized data publication across organizational boundaries [12].

More broadly, European data space initiatives emphasize federated trust, identity, and governance mechanisms to enable cross-organizational data exchange while preserving sovereignty, aligning with the boundary-oriented role Solid plays in MuMo [11, 18].

Rather than enforcing a single global schema or repository, MuMo enables institutions to publish data independently and allows consumers to integrate only what they need, when they need it. This approach proved compatible with legacy systems and institutional autonomy, both of which are common in Digital Humanities settings.

Lesson learned: Dataspace architectures are particularly well suited to DH environments where data is distributed, governance is decentralized, and collaboration is episodic.

6. Impact for the Digital Humanities Community

Many Digital Humanities projects rely on data that is collected continuously over long periods of time and embedded in local infrastructures, such as sensor data, annotations, or evolving metadata. In practice, such data often remains siloed within project-specific systems, limiting its reuse, comparability, and shareability across institutional boundaries.

Digital Humanities researchers have increasingly emphasized that the sustainability of digital resources is not only a technical issue but also an institutional and socio-technical one, shaped by funding horizons, maintenance practices, and governance constraints [20, 6].

The experience reported in this paper suggests that **dataspace-oriented approaches** are well suited to such settings. By allowing data to remain under the control of the institution that produces it, while enabling integration at the point of use, dataspaces provide a viable alternative to centralized platforms. This is particularly relevant for DH collaborations involving multiple partners with heterogeneous governance structures and long-lived legacy systems.

The use of **semantic representations as infrastructural connectors**, rather than as expressive modeling exercises, further supports interoperability in constrained environments. Lightweight semantic alignment enables independently managed datasets to be combined, extended, and reinterpreted over time, without requiring uniform tooling or deep semantic expertise from end users.

Access control emerged as a critical dimension of data reuse. Rather than pursuing maximal technical expressiveness, the project highlights the importance of aligning authorization mechanisms with existing organizational practices. Group-based access control, when grounded in curatorial workflows, can provide sufficient protection and flexibility while remaining understandable and manageable in daily operation.

Finally, breaking down data silos enables new forms of longitudinal and comparative analysis that are difficult to achieve with isolated systems. The ability to combine data from multiple independent sources—under controlled access conditions—opens opportunities for richer contextualization and collaboration in a wide range of Digital Humanities scenarios.

Together, these observations point toward a pragmatic path for DH infrastructures that emphasizes decentralization, semantic interoperability, and governance-aware design over comprehensive but fragile integration solutions.

7. Conclusion and Future Work

This paper reported on MuMo (Museum Monitoring), a three-year applied research project that explored the use of dataspace principles for environmental monitoring in museum practice. By integrating semantic data publication, event-based access, and decentralized governance with existing monitoring infrastructure, MuMo addressed practical challenges related to long-running data collection, cross-institutional collaboration, and controlled data sharing.

A central outcome of the project is the demonstration that dataspace-oriented architectures can be deployed incrementally alongside legacy systems. Rather than requiring centralized platforms or uniform tooling, MuMo shows how interoperability and data reuse can be achieved while preserving institutional autonomy and established workflows. The project further illustrates that design choices grounded in curatorial practice—such as group-based access control—can be both sufficient in practice and easier to sustain than more fine-grained alternatives.

Several limitations remain. Access control is currently restricted to group-level permissions, reflecting the expressive capabilities of the legacy dashboard rather than technical constraints of the dataspace architecture. Future work could explore finer-grained authorization mechanisms, such as temporal or measurement-level access, should corresponding user requirements arise. In addition, while the system supports aggregation across multiple MuMo deployments, integrating external data sources beyond the project has not yet been explored in deployed settings.

In conclusion, MuMo provides an in-use perspective on how Solid-based data governance, Linked Data Event Streams, and lightweight semantic modeling can support practical Digital Humanities workflows. The project highlights the value of aligning technical architectures with organizational realities and contributes concrete lessons for the design of interoperable, sustainable DH data infrastructures.

References

[1] Raf Buyle, Laurens De Vocht, Mathias Van Compernolle, Dieter De Paepe, Ruben Verborgh, Ziggy Vanlishout, Bjórn De Vidts, Peter Mechant, and Erik Mannens. Oslo: Open stan-

dards for linked organizations. In *Proceedings of the international conference on electronic governance and open society: Challenges in Eurasia*, pages 126–134, 2016.

[2] Sarven Capadisli, Tim Berners-Lee, Ruben Verborgh, and Kjetil Kjernsmo. Solid protocol, December 2022. URL https://solidproject.org/TR/2022/protocol-20221231.

[3] Story Henry Capadisli Sarven and Tim Berners-Lee. eb access control (wac). Technical report, MIT CSAIL, 2024. URL https://solidproject.org/TR/wac.

[4] Michael Compton, Payam Barnaghi, Luis Bermudez, Raul Garcia-Castro, Oscar Corcho, Simon Cox, John Graybeal, Manfred Hauswirth, Cory Henson, Arthur Herzog, et al. The ssn ontology of the w3c semantic sensor network incubator group. *Journal of Web Semantics*, 17:25–32, 2012.

[5] European Commission. Linked data event streams (ldes) | semic support centre | interoperable europe portal, 2026. URL https://interoperable-europe.ec.europa.eu/collection/semic-support-centre/linked-data-event-streams-ldes. Accessed: 2026-01-20.

[6] Katrina Simone Fenlon. Sustaining digital humanities collections: challenges and community-centred strategies. *International Journal of Digital Curation*, 15(1):13–13, 2020.

[7] Michael Franklin, Alon Halevy, and David Maier. From databases to dataspaces: a new abstraction for information management. *ACM Sigmod Record*, 34(4):27–33, 2005.

[8] Alon Halevy, Michael Franklin, and David Maier. Principles of dataspace systems. In *Proceedings of the twenty-fifth ACM SIGMOD-SIGACT-SIGART symposium on Principles of database systems*, pages 1–9, 2006.

[9] Armin Haller, Krzysztof Janowicz, Simon J. D. Cox, Maxime Lefrançois, Kerry Taylor, Danh Le Phuoc, Joshua Lieberman, Raúl García-Castro, and Claus Stadler. Semantic sensor network ontology. W3c recommendation, W3C, 2017. URL https://www.w3.org/TR/vocab-ssn/.

[10] Vincent C Hu, David Ferraiolo, Rick Kuhn, Adam Schnitzer, Kenneth Sandlin, Robert Miller, Karen Scarfone, et al. Guide to attribute based access control (abac) definition and considerations. *NIST special publication*, 800(162):1–54, 2014.

[11] International Data Spaces Association. Ids reference architecture model - IDS-RAM 4.0. Technical report, International Data Spaces Association, feb 2023. URL https://docs.internationaldataspaces.org. Accessed: [Insert Date Here].

[12] Dwight Van Lancker, Steven Logghe, Julián Andrés Rojas, Annelies De Craene, Ziggy Vanlishout, and Pieter Colpaert. Semantic and technically interoperable data exchange in the flanders smart data space. In Gianluca Demartini, Katja Hose, Maribel Acosta, Matteo Palmonari, Gong Cheng, Hala Skaf-Molli, Nicolas Ferranti, Daniel Hernández, and Aidan Hogan, editors, *The Semantic Web – ISWC 2024*, pages 289–303, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-77847-6.

[13] Essam Mansour, Andrei Vlad Sambra, Sandro Hawke, Maged Zereba, Sarven Capadisli, Abdurrahman Ghanem, Ashraf Aboulnaga, and Tim Berners-Lee. Solid: A platform for decentralized social applications based on linked data. Technical report, MIT CSAIL, 2016. URL https://solidproject.org/TR/solid.

[14] Stefan Michalski. The ideal climate, risk management, the ashrae chapter, proofed fluctuations, and towards a full risk analysis model. *Experts roundtable on sustainable climate management strategies*, pages 1–19, 2007.

[15] Laura M Rodríguez Peralta and Lina M Brito. An integrating platform for environmental monitoring in museums based on wireless sensor networks. *International Journal on Advances in Networks and Services*, 3(1/2):114–124, 2010.

[16] Dian Marie Ross. *DIGITAL TWINS, HARDWARE ORACLES AND DISTRIBUTED LEDGER TECHNOLOGIES*. PhD thesis, UNIVERSITY OF BRITISH COLUMBIA (Vancouver), 2024.

[17] Wout Slabbinck, Ruben Dedecker, Sindhu Vasireddy, Ruben Verborgh, and Pieter Colpaert. Linked data event streams in solid ldp containers. In *Managing the Evolution and Preservation of the Data Web 2022*, volume 3339, pages 28–35. CEUR, 2023.

[18] Hubert Tardieu. *Role of Gaia-X in the European Data Space Ecosystem*, pages 41–59. 07 2022. ISBN 978-3-030-93974-8. doi: 10.1007/978-3-030-93975-5_4.

[19] Lefteris Tsipi, Demosthenes Vouyioukas, Georgios Loumos, Antonios Kargas, and Dimitrios Varoutas. Digital repository as a service (d-raas): Enhancing access and preservation of cultural heritage artifacts. *Heritage*, 6(10):6881–6900, 2023.

[20] Joanna Tucker. Facing the challenge of digital sustainability as humanities researchers. *Journal of the British Academy*, 10:93–120, 2022.

[21] Dwight Van Lancker, Pieter Colpaert, Harm Delva, Brecht Van de Vyvere, Julián Rojas Meléndez, Ruben Dedecker, Philippe Michiels, Raf Buyle, Annelies De Craene, and Ruben Verborgh. Publishing base registries as linked data event streams. In *International Conference on Web Engineering*, pages 28–36. Springer, 2021.

[22] Verstraete, Melanie and Verbrugge, Sofie and Colle, Didier. Solid : enabler of decentralized, digital platforms ecosystems. In *ITS 31st European Conference 2022, Proceedings*, page 19, 2022. URL https://itseurope.org/2022/.